

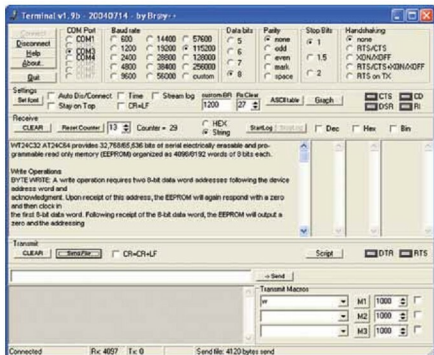


Fig. 6: Terminal v1.9b output window

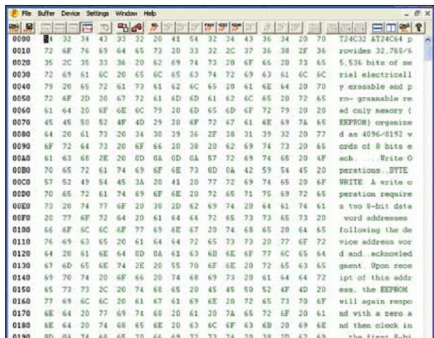


Fig. 7: TopView output text captured from EEPROM device

the platform window and COM port (say, COM1) in the port window. Next, select ATmega2560 as 'Device' and ISP as 'Programming Mode' and browse your project hex file from Project→Debug→Exe folder. Press 'Program' button on the STK500 dia-

logue box to burn the hex file into your microcontroller.

Boot loader implementation greatly simplifies your work by eliminating the need of a separate programmer again and again. Bootloader code is provided in this month's EFY-CD.

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - ATmega2560 microcontroller
- IC3 - MAX232 RS-232 driver
- IC4 - AT26C64 EEPROM
- LED1-LED3 - 5mm light-emitting diode

Resistors (all 1/4-watt, ±5% carbon):

- R1-R3 - 470-ohm
- R4, R5 - 4.7-kilo-ohm
- R6, R7 - 10-kilo-ohm

Capacitors:

- C1 - 100µF, 25V electrolytic
- C2 - 0.1µF ceramic
- C3, C4 - 22pF ceramic
- C5-C8 - 1µF, 16V electrolytic
- C9 - 10µF, 16V electrolytic

Miscellaneous:

- CON1 - DC connector
- CON2 - ISP 6-pin male connector
- CON3 - 9-pin D-type female connector
- S1, S2 - Tactile switch

You can burn boot loader code into MCU flash memory using STK500 or any other inexpensive programmer such as Serial Port AVR-01 Programmer from http://www.onlinetps.com/shop/index.php?main_page=index&cPath=65

You can now program the MCU in your application board using software such as AVR-OSP II.

Using AVR-OSP II and AVR application board. AVR-OSP II is an Open Source programmer tool for AVR chips. Download its Version 547 from www.esnips.com/doc/ad-233fbc-7a98-4ebc-9b9c-d2b9f4d5f786/. Unzip the contents to a single folder and run AvrOspIi.exe. In 'Configurations' tab, set bits per second as 19,200 and select COM port. Now select the protocol as AVR11.

Press switch S2 and reset switch S1 simultaneously. While releasing S1, keep S2 pressed. LED2 connected to pin 41 (PL6) of ATmega2560 should glow to indicate that the board is in programming mode. Release switch S2.

Press 'Auto Detect' in 'Program' window. It should detect the AVR board ATmega2560 on it. Now select the hex file that has been created by your microcontroller code. Enable